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"creator" : "ari.neko@cvut.cz",
"timestamp" : "09-07-2026_17:43:52",
"variant" : 1,
"subjectName" : "mathematics",
"categories" : ["Combinatorics", "Planimetry", "Mathematical Logic"],
"questions" : [{
"question" : "How many distinct subsets can be formed from a set containing 5 elements?",
"answers" : ["32", "16", "25", "64"],
"correctIndex" : 0
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"question" : "What is the value of the combinatorial coefficient C(5, 2)?",
"answers" : ["20", "15", "10", "5"],
"correctIndex" : 2
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"question" : "Which principle is also known as the Dirichlet Box Principle?",
"answers" : ["Bijection Principle", "Pigeonhole Principle", "Multiplication Principle", "Inclusion-Exclusion Principle"],
"correctIndex" : 1
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"question" : "How many unique permutations can be made from the letters in the word 'APP'?",
"answers" : ["1", "9", "3", "6"],
"correctIndex" : 2
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"question" : "According to Brahmagupta's formula, what does it calculate?",
"answers" : ["Volume of a pyramid", "Perimeter of an ellipse", "Area of a scalene triangle", "Area of a cyclic quadrilateral"],
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"question" : "What is the area of a circle with a diameter of 10?",
"answers" : ["100 * pi", "25 * pi", "10 * pi", "50 * pi"],
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"answers" : ["Contingency", "Tautology", "Contradiction", "Fallacy"],
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"question" : "What is the negation of the statement 'All primes are odd'?",
"answers" : ["Some primes are odd", "There exists at least one prime that is not odd", "No primes are odd", "All primes are even"],
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"question" : "Which logical connective represents an 'exclusive or' (XOR)?",
"answers" : ["Both inputs are true", "Exactly one input is true", "At least one input is true", "Both inputs are false"],
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"question" : "What logic law simplifies 'not (A and B)' into '(not A) or (not B)'?",
"answers" : ["Distributive Law", "Associative Law", "De Morgan's Law", "Commutative Law"],
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